INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, M99 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M99 Cartridge Actuated Initiator Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-2550	-	Ammunition, General Specification For
		NOTE: MIL-A-2550 is inactive for new design and will be eventually replaced.
MIL-DTL-60525	-	Initiator, Cartridge Actuated, M99 Metal Parts Assembly
MIL-I-82767	-	Initiator, Cartridge Actuated, M99 Remanufactured Metal Parts Assembly
MS19060	-	Balls, Bearing, Corrosion Resistant Steel
MS28741	-	Hose Assembly, Detachable End Fitting, Medium Pressure

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NSWC IHD (CAGE CODE 19200 and 19203) DRAWINGS

8597230	-	Cartridge, Impulse, M91 Assembly

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1377

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8830984 - Box, Fiberboard (Inner) Packing, Ammunition, for Initiator, Cartridge Actuated, M27, M86, M87, M98, M99 or XM107

8830985 - Box, Packing, Ammunition for Initiator, Cartridge Actuated, M27, M86, M87, M98, M99 or XM107

9241155 - Box, Fiberboard (Outer) Alternative Packing, Ammunition, for Initiator, Cartridge Actuated, M27, M86, M87, M98, M99 or XM107

10520405 - Initiator, Cartridge Actuated, M99 Assembly

10520407 - Chamber, Initiator

10520409 - Pin, Initiator

10520410 - Housing, Firing Pin

10520411 - Pin, Firing

10520412 - Spring, Initiator

10520414 - Filter

10520415 - Seal

10520496 - "O" Ring

(Copies of these documents are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquires, or problems to nswc_iheodtd_dmo@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS-832 - Cap, Protective, Flared Fitting (Plastic)

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM E1742/E1742M - Standard Practice for Radiographic Examination

(Copies of this document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the initiators shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the Technical Design Authority prior to manufacture.

3.2 Primary components. For the purpose of this specification, the M91 impulse cartridge (Drawing 8597230) is defined as a primary component. Only primary components from a single lot shall be used in a production lot of initiators. One primary component production lot may be used in more than one initiator production lot.

3.3 Component parts. All component parts shall meet the requirements of MIL-DTL-60525 or MIL-I-82767 prior to assembly.

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3.4 Radiographic examination. Each initiator assembly shall be free of defects when radiographically examined as specified in 4.4.2.

3.5 Breakaway torque. The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation (see 4.4.3).

3.6 Residual magnetism. The initiator shall not deflect the indicator of the compass more than 5 degrees in either direction using the procedure specified in 4.4.4.

3.7 Firing pin release. The firing pin of the assembled initiator, less safety pin, cartridge, and chamber shall release from the cocked position when the application of an axial load of 18 to 25 pounds has withdrawn the initiator pin 0.370 inch minimum and 0.425 inch maximum (see 4.4.5).

3.8 Initiator pin security. Application of an axial load of 200 to 220 pounds to the initiator pin of the assembled initiator shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin (see 4.4.6).

3.9 Functional.

3.9.1 Ballistics. Initiators loaded with cartridges from a single accepted lot and temperature-conditioned shall perform within the pressure values and ignition delay time (see 6.5.1) specified in table I. Pressure values shall be recorded at a pressure chamber with a volume of 1 cubic inch attached to the end of 15 feet of aircraft hose, assembly number MS28741-4.

TABLE I. Pressure values.

Temperature (°F)	Pressure (psi)		Ignition delay time, maximum (milliseconds) (see 6.5.1)
	Minimum	Maximum	
-60 to -70	500	1,425	65
65 to 75	600	1,550	65
195 to 205	700	1,700	65

3.9.2 Misfires. There shall be no misfires.

3.9.3 Mechanical failure. The initiator shall function without mechanical failure (see 6.5.2).

3.10 Workmanship. Initiators shall be free of cracks, splits, incomplete or damaged threads on outlet ends, or other defects that might prevent proper mating with connecting parts or proper functioning of the device.

3.10.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.10.2 Burr. No part shall have a burr that might interfere with the assembly or function of the item or that might be injurious to personnel handling the item.

3.10.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Production inspection (see 4.2).

b. Lot acceptance inspection (see 4.3).

4.1.1 Inspection. Inspection shall be in accordance with MIL-A-2550.

4.2 Production inspection. All initiators shall be inspected and screened to the examinations and tests listed in table II. Any initiator that fails to meet the requirements of table II shall be rejected and removed from the lot.

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TABLE II. Production inspection.

Component	Test	Requirement	Verification
All components	Inspection	3.3	4.4.1
Initiator	Radiographic examination	3.4	4.4.2
Cap and chamber assembly	Breakaway torque	3.5	4.4.3
Initiator	Residual magnetism	3.6	4.4.4
Firing pin assembly	Firing pin release	3.7	4.4.5
Initiator pin assembly	Initiator pin security	3.8	4.4.6
All components	Workmanship	3.10	4.4.1, 4.4.2

4.3 Lot acceptance inspection. Lot acceptance inspection shall consist of the examinations and tests in table III. Failure of any sample to comply with the requirements of table III shall be cause for rejection of the lot represented.

TABLE III. Lot acceptance inspection.

Test/inspections	Requirement	Test
Ballistic (-65±5 °F)	3.9.1	4.4.7.1
Ballistic (70±5 °F)	3.9.1	4.4.7.1
Ballistic (200±5 °F)	3.9.1	4.4.7.1

4.3.1 Lot. Unless otherwise specified (see 6.2), formation of lots and presentation of lots shall be in accordance with ASQ/ANSI Z1.4. A lot shall consist of an identifiable group of assembled initiators containing metal parts assemblies produced by a homogenous process and loaded with one lot of accepted cartridges.

4.3.2 Sampling. A random sample of 21 initiators, including the samples for investigative purposes, shall be selected from the production lot.

4.4 Examinations and tests.

4.4.1 Component part examination. One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects (see table IV). Four power (4X) magnification may be used for inspection of defects such as burr and foreign matter. When compliance with the applicable requirement is in doubt because of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

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TABLE IV. Classification of defects.

Cartridge, Impulse, M91, Drawing 8597230, covering a detail of Drawing 10520405		
Characteristic	**Defect**	**Method of Inspection**
Critical		
1	Mixed cartridges	Visual
2	Metal defective	Visual
3	Marking missing, incorrect, or illegible	Visual
Major		
101	Foreign matter present	Visual
Minor		
201	Case scratched, wrinkled, bulged, or dented	Visual
202	Varnish missing or inadequate	Visual
Assembly, Drawing 10520405, less all Drawings except 10520410, 10520411, and MS19060-1008		
Critical		
1	Protrusion of firing pin	Gage
2	Steel ball missing	Gage
3	Striker hole in firing pin housing obstructed	Visual
Major		
101	Burr	Visual
102	Foreign matter present	Visual
Minor		
None defined		
Assembly, Drawing 10520405, less Drawings 8597230, 10520407, 10520496, 10520414, and NAS-832-4		
Critical		
1	Firing pin not cocked	Visual/Manual
2	Any parts missing	Visual
3	Improper assembly	Visual
Major		
101	Safety pin loose or deformed	Visual
102	Burr	Visual
103	Foreign matter present	Visual
Minor		
None defined		

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TABLE IV. Classification of defects – Continued.

Assembly, Drawing 10520405, less all Drawings except 10520407, 10520414, and 8597230		
Critical		
1	Cartridges not properly seated	Visual
2	Filter missing or insecure	Visual
3	Nylon patch missing or improperly applied	Visual
Major		
101	Burr	Visual
102	Foreign matter present	Visual
Minor		
None defined		
Assembly, Drawing 10520405		
Critical		
1	Cartridge face not seated against base of firing pin housing	X-ray
2	Any part missing except as otherwise classified	Visual/X-ray
3	Improper assembly	Visual/X-ray
4	Marking missing, incomplete, or illegible	Visual
Major		
101	Set screw missing or insecure	Visual
102	Burr	Visual
103	Foreign matter present	Visual
Minor		
201	Thread sealing compound missing from set screws	Visual
202	Protective cap missing	Visual
203	Protective finish inadequate or defective	Visual
Box, Fiberboard (Inner) Ammunition Packing Drawing 8830984		
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Box damaged	Visual
203	Sealing strip missing, improperly applied	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Cut, damaged, or improperly sealed bag	Visual

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TABLE IV. Classification of defects – Continued.

Box, Ammunition Packing, Drawing 8830985		
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Strapping missing, loose, broken, or improperly assembled	Manual/Visual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units, incorrect	Visual
Box, Fiberboard (Outer) Alternative Ammunition Packing, Drawing 9241155		
Critical		
None defined		
Major		
None defined		
Minor		
201	Contents loose	Manual
202	Box broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.4.2 Radiographic examination. All assembled initiators shall be radiographically examined in accordance with ASTM E1742/E1742M, with exception that an initiator assembly shall be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The initiators shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All initiators shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive numerical order in each production lot. The initiators shall be arranged on boards or trays in consecutive order and each radiograph shall carry a permanent identification of the units displayed thereon. Any missing serial numbers are to be identified on the X-rays. The radiographic plate identification shall include item nomenclature, the complete lot number, the span of serial numbers displayed, and the contract number. Defective initiators found during X-ray review shall be clearly identified on the radiographic plate and those defective items shall be removed from the production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

a. O-ring present P/N 10520496
b. Steel balls present (3 ea) P/N MS19060-1008
c. Firing pin P/N 10520411
d. Spring, initiator P/N 10520412
e. Seal P/N 10520415

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f. Firing pin housing P/N 10520410
g. Filter P/N 10520414
h. Cartridge, Impulse M91 P/N 8597230
i. Initiator pin P/N 10520409

4.4.3 Breakaway torque. Each initiator shall be subjected to the breakaway torque test of the threaded joint between the cap and the chamber. A torque wrench set at the torque requirement specified in 3.5 shall be used in conjunction with a fixture to determine the adequacy of the joints. The chamber shall be secured in a vise, and the torque shall be applied to the cap. Failure of any initiator to comply with the requirements of 3.5 shall be cause for rejection of the initiator.

4.4.4 Residual magnetism. Each initiator shall be subjected to the test for excessive residual magnetism. The initiator shall be tested for excessive residual magnetism using a calibrated compass in an area free of local magnetic effects by placing the assembly 5 inches from the end at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The initiator shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90-degree rotation of the assembly for four full passes. Failure of any initiator to comply with the requirement of 3.6 shall be cause for rejection of the initiator.

4.4.5 Firing pin release. The initiator pin of each subassembly, consisting of all parts, except the safety pin, cartridge, and chamber, shall be subjected to the firing-pin release test. The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction, in conjunction with the fixture that registers the amount of pull. To achieve the pull force requirement of 18 to 25 pounds specified in 3.7, selective reassembly may be necessary (i.e., changing initiator pins or springs) for those assemblies that are out of range. Failure of any subassembly to comply with the requirements of 3.7 shall be cause for rejection of the subassembly.

4.4.6 Initiator pin security. The initiator pin of each subassembly consisting of all parts, except the chamber and cartridge, shall be subjected to the initiator-pin security test. The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with a fixture which registers the amount of pull. The test shall be performed with the safety pin in both positions. Failure of any subassembly to comply with the requirements of 3.8 shall be cause for rejection of the subassembly.

4.4.7 Functional tests.

4.4.7.1 Ballistics. Twenty-one units shall be tested to determine compliance with the requirements of 3.9.1. The units shall be temperature-conditioned as provided in table V. Each assembled initiator shall be conditioned for a period of time not less than 4 hours nor greater than 24 hours. The test sample shall be rigidly supported in the fixture and fired. The sample shall be tested within 5 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned. When reconditioning initiators at -60 °F, all condensation shall be removed from the interior surfaces before returning the initiator to the conditioning chamber. Required pressure-time readings shall be taken as specified in 3.9.1. Failure of any unit to comply with the requirements of 3.9.1 shall be cause for rejection of the lot.

TABLE V. Conditioning temperature.

Temperature (°F)	Sample size
-60 to -70	7
65 to 75	7
195 to 205	7

4.4.7.2 Retest. There shall be no retests except as specified in 4.4.7.3.

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4.4.7.3 Test failure. If a test failure is attributable to an assignable cause, excluding the test initiator, the original test results shall be discarded and that test reconducted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in aircrew emergency escape systems. Since the initiator assembly was designed for use in military aircraft, there is no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Production test activity and lot size (see 4.3.1).

c. Direction for shipping radiographs of entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.4.2).

d. Packaging requirements (see 5.1 and 6.3).

e. Safety precautions (see 6.4).

6.3 Suggested packaging. Packaging in accordance with SPI F01-271-7161.

6.4 Safety precautions. The safety precaution requirements of DoD Manual 4145.26, DoD Contractor's Safety Manual For Ammunition and Explosives, are applicable and should be as specified (see 6.2) as required by the Federal Acquisition Regulation (FAR), Subpart 23.3. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of OP 5, Ammunition and Explosives Safety Ashore, should be made applicable.

6.5 Definitions.

6.5.1 Ignition delay time. Ignition delay time is defined as the time from primer impact to the start of pressure rise.

6.5.2 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.6 Subject term (key word) listing.

Ejection systems, aircrew

Emergency escape systems, aircrew

Escape systems, aircrew

Life-saving systems, aircrew

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Navy – OS
Air Force – 99

Preparing activity:
Navy – OS
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